جامعة خليفة
Khalifa University

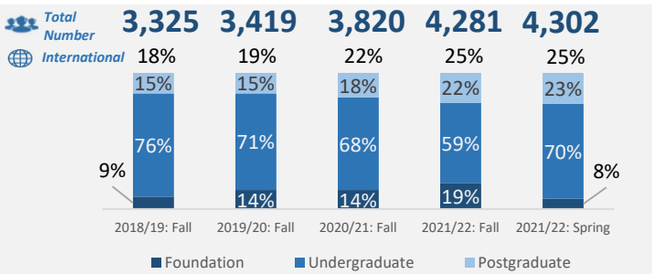
Khalifa University

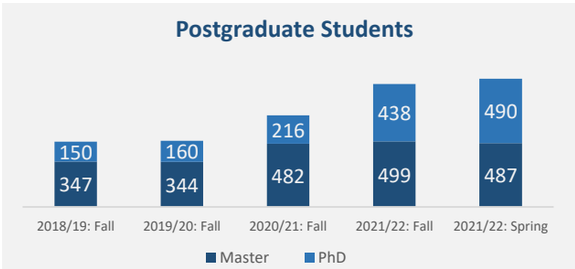
Strategic Plan (2021-2031)

Khalifa University in Numbers

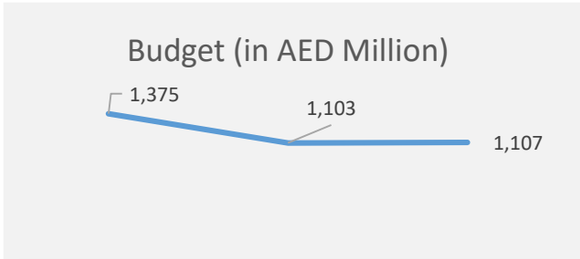
College of Arts and Sciences | **College of Engineering** | **College of Medicine and Health Sciences** | **3 Institutes** | **20 Research Centers**

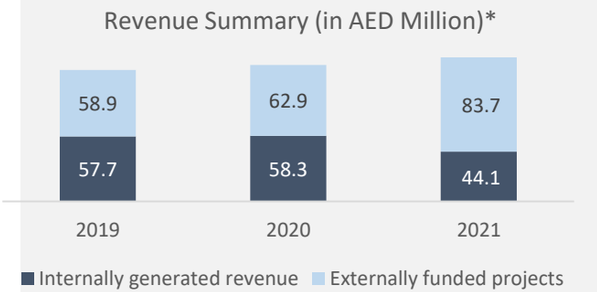
Student Enrollment



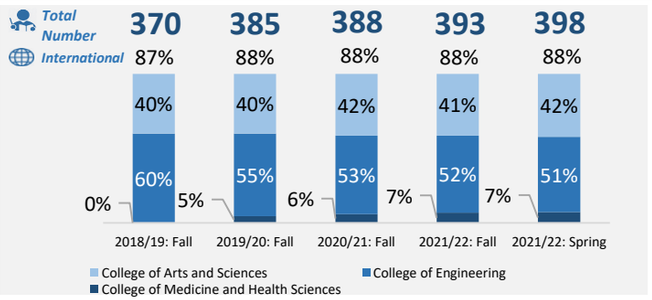


Finance





Faculty



Year	No. of Publications Per Faculty with PhD	% of Faculty with Doctoral Degree
2018	4.1	87%
2019	4.1	90%
2020	4.3	91%
2021	4.4	92%
2022	4.7	92%

Accredited Programs By CAA

Year	Undergraduate	Postgraduate
2018	12	16
2019	13	18
2020	14	22
2021	16	26
2022	17	27

Research Productivity

Source: Scival

Year	No. of Publications	% of International Collaboration
2018	1,322	65.7%
2019	1,420	70.3%
2020	1,513	73.8%
2021	1,710	77.3%

Patents per year

Year	Filed	Issued
2018	84	31
2019	77	51
2020	101	36
2021	108	32

Total Issued
234

* Internally generated revenue: Actual basis (cash) which is transferred to DoF; Externally funded projects: Accrual basis (billing only) not cash basis. Maintained in KU ledger for externally funded research sponsorship projects and other projects funded by external parties.

Strategy map for Khalifa University of Science and Technology

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KU STRATEGY MAP 2021-2031

Vision

To be a catalyst to the growth of Abu Dhabi and the UAE's rapidly developing knowledge economy, an education destination of choice, and a global leader among research intensive universities of the 21st century

Emirate Programs	Prepare students to compete in the labor market	Provide a stimulating and supportive research, science and technology environment	Enhance innovation system and culture
KU Strategic Priorities	**1- World Class Education** Prepare future leaders to an internationally recognized standard.	**2- Influential Research** Produce world-class research with local relevance and international impact	**3- Catalyst for Economic Development** Enrich the national economy through innovation and research commercialization
Initiatives	I1 Promote educational excellence in all programs I2 Cultivate a vibrant environment for research and innovation I3 Work with a student focus I4 Enhance the university reputation both locally and internationally I5 Diversify financial resources I6 Foster engagement and satisfaction in the university I7 Optimize efficiency and effectiveness of processes		
Values	Excellence Leadership	Innovation Community	Integrity Life Long Learning Ambition Mutual Respect

Summary

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Strategic Priorities

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Initiatives

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Objectives/Targets

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Key Strategic targets

Khalifa University of Science and Technology mission

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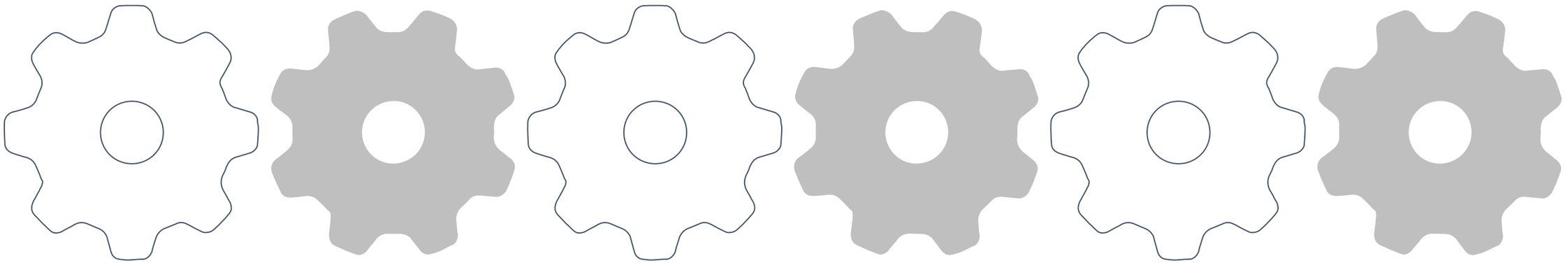

As a world-class, research-intensive institution, Khalifa University of Science and Technology will:

- Set new standards in education, research and scholarship that will benefit the UAE and the world.
- Drive Abu Dhabi and the UAE as a knowledge destination and engine for socioeconomic growth through active translation of research into the nation's economy.
- Seamlessly integrate research and education to produce world leaders and critical thinkers in applied science, engineering, management and medicine.
- Continuously innovate and integrate the global standard in methods of learning and discovery.
- Build a diverse community of service-oriented, ambitious and talented individuals, through an environment that encourages and nurtures creative inquiry, critical thinking, and human values.
- Empower the community with practical and social skills, business acumen and a capability for lifetime learning that will enrich the workforce of the country.

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Strategic objectives and targets

KU has set 5 Key strategic targets to be achieved by 2031

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1 Graduates Employment Rate

80%-90%

	2017/18	2018/19	2019/20
Total	77%	76%	68%*
Nationals	77%	81%	66%*

2 Publications are in the top 5% cited papers per year

20-25%

2018	2019	2020	2021
8.1%	10.8%	11.4%	16.2%

3 International Ranking

Top 80-100

2018	2019	2020	2021	2022
QS 451-460	QS 315	QS 268	QS 211	QS 183

4 Number of Start ups **

14-20

2017	2021
1 Start Up	9 KU Start ups

5 Diversified Funding Sources

20-25%

2018	2019	2020	2021
10.5%	8%	10.2%	10.4%

% of the university revenue from external funding sources from total revenue

* COVID – 19 and the reduction of the number of students who are externally sponsored by industry have impacted the employment rates

** Start-ups target: 3-5 Start-ups (Valuation ~AED 5 M), 3-5 Start-ups (Valuation ~AED 2.5 M), 8 - 10 Start-ups (Valuation ~AED 1 M)

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KU: Initiatives, Objectives and Targets 2021-2031

Priority 1: Prepare future leaders to an internationally recognized standard

Strategic Initiatives	Objectives	Key Performance Indicators	Baseline Data 2020	Targets	Year
I1 Promote educational excellence in all programs	Widen access to students with outstanding academic potential	Total students enrolment	3820*	4500-5000	2031
	Align academic programs with the local and global labor market needs	% of new established programs aligned to the global and local market needs starting from 2021	100%	100%	2024
	Develop a new learning outcomes assessment framework at the course, programs, and institutional levels	% of assessed programs learning outcomes	50%	90%-100%	2024
	Develop holistic experiential learning programs	Number of experiential learning programs a student should attend before graduating	1	1-2	2024
	Develop a career guidance system	% of students who are engaged in institutionalized career guidance sessions	61%**	70%-90%	2024
I2 Work with a student-centric focus	Develop the university innovation and entrepreneurship system	Completion percentage of the innovation and entrepreneurship ecosystem	KIC and embedded entrepreneurship and innovation courses	80%-100%	2024
	Develop a life-long culture of community service beyond the classroom	Number of community services activities	377	350-400	2031
	Establish a mobility program for students, faculty, researchers and staff	% of the students and % of the faculty participate in exchange programs annually	0.79 - 0 %	5%-5%	2031

Strategic Plan Key Targets

Year 2031

80% -90% employment rate

* Fall 2020
** Includes career guidance sessions in Summer 2021 as of June 24, 2021

Priority 2: Produce world-class research with local relevance and international impact

Strategic Initiatives	Objectives	Key Performance Indicators	Baseline Data 2020	Targets	Year
I3 Cultivate a vibrant environment for research and innovation	Align the postgraduates programs with the research strategy	% of aligned postgraduate programs and research strategy	70%	70%-90%	2024
	Develop a comprehensive competitive internal research funding program	% of faculty received internal research funds	70%	70%-90%	2024
	Diversify and increase external research funds	% of research expenditure from external sources out of total expenditure on research	38%	40%-60%	2025
	Strengthen research ethical culture framework across the university	% of the university staff attended training programs on research ethics	89%	90%-100%	2022
	Review and revise as needed the research strategic plan	Completion percentage of implementing the updated research strategic plan	The strategy Developed and approved	80%-100%	2024
	Attract and organize international conferences and summits	Number of conferences/summits per year	3	4-5	2024
	Increase number of postgraduate students	% of postgraduate students from the total student enrolment	21.2% 14.7% MSc 6.6% PhD (excluding Prep)	25% 12.5% MSc 12.5% PhD	2031

Strategic Plan Key Targets

Year 2031

12%-15% of the publications are in the top 5% cited papers

Priority 2: Produce world-class research with local relevance and international impact

Strategic Initiatives	Objectives	Key Performance Indicators	Baseline Data 2020	Targets	Year
I4 Enhance the university reputation both locally and internationally	Develop a multidimensional program to attract and retain diverse outstanding students	% international students	25% (excluding PP students)	30%-35%	2031
	Recruit and retain high caliber faculty, researchers and staff	% of the newly recruited faculty with h-5 index >= 4	64%	80%-90%	2024
	Realize a sustainable , healthy and safe campus environment for all community members	% compatibility with the national standards	90%	80% -90%	2024
	Enhance the communication plan and outreach activities	Completion percentage of the updated and enhanced communication and marketing plan	Communication and Marketing Plan	80%-100%	2024
	Increase publications in journals with high impact factor	Average number of publications indexed in Scopus per faculty	4.3	4-5	2024
	Increase filed and issued patents in USPTO and PCT	Number of annual registered patents	32	30 - 40	2024
	Build partnerships with local and international universities	Number of new strategic partnerships with international universities/per year*	31	2 - 4	2024

Strategic Plan Key Targets

Year 2031

Top 80-100 ranked universities

* Active partnerships may result in joint research projects and minds exchange

Priority 3: Enrich the national economy through innovation and research commercialization

Strategic Initiatives	Objectives	Key Performance Indicators	Baseline Data 2020	Targets	Year	Strategic Plan Key Targets
I5 Diversify financial resources	Develop competitive fellowship programs for students	% of sponsored students by external entities	15%	25% - 30%	2031	Year 2031
	Establish and manage start-up companies	Total number of start-up companies that have spun out of KU projects/research	2 Start ups using KU licensed IPs 2 Start ups currently in the Catalyst Incubator	15 - 20	2031	Diversified Funding Sources 20%-25%
	Create diversified financial resources for self sustaining university	% of university revenue from external funding sources out of total revenue	10.2%	20% - 25%	2031	Year 2031
I6 Foster engagement and satisfaction in the university	Design and implement professional development programs for researchers, faculty, and staff to contribute to the university strategic plan	% of the university human capital who received documented and structured training	40%	80%-100%	2024	15- 20 startups

Priority 3: Enrich the national economy through innovation and research commercialization

Strategic Initiatives	Objectives	Key Performance Indicators	Baseline data 2020	Targets	Year	Strategic Plan Key Targets
I7 Optimize efficiency and effectiveness of processes	Ensure the university sustainability	% of employed nationals out of the total number of employees*	24%	30%-35%	2031	
	Utilize state-of-the-art technologies and facilities in an optimal and efficient manner	Utilization rate of the university facilities	80%	80%-90%	2024	
	Promote a working culture of excellence and accountability	Completion percentage of the Process excellence framework	50%	80%-100%	2024	
	Provide reliable, accurate and timely reports to different stakeholders	Completion percentage of the holistic university information system**	70%	80%-100%	2024	
	Develop and implement of performance management system across the university	Completion percentage of the automated performance management system	70%	80%-100%	2022	

* Includes faculty and admin staff
** ~60% of the university data is currently available in the 3 management information systems

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Research Priorities

Research - Focus topics

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Clean and Renewable Energy
- Solar and other renewable energy
- Electrical power transmission and distribution
- Energy storage
- Energy efficient cities and buildings
- Nuclear energy

Water and Environment
- Desalination
- Waste treatment and utilization (wastewater, industrial and municipal waste streams)
- Environmental resource management

Hydrocarbon Exploration and Production
- Reservoir characterization, evaluation and surveillance
- Enhanced/improved oil recovery
- Gas processes
- Asset integrity
- Drilling and completion

Healthcare
- Genomics and Bioinformatics
- Medical imaging
- Biomedical robotics and devices
- Bio-materials and cellular mechanics
- Healthcare management

Aerospace (Aero/Astro)
- Cubesats and nanosatellites
- Remote sensing and modeling
- *Space sciences*

Supply Chain and Logistics
- Transportation
- Manufacturing logistics
- Supply-chain strategy
- Maintenance repair and overhaul (MRO)

Advanced Materials and Manufacturing
- Computational materials engineering
- Composites
- Enhanced materials systems (nanomaterials and nanocomposites, multifunctional materials, smart materials, metamaterials)
- Biomaterials
- Additive manufacturing
- Manufacturing automation

Robotics, Artificial Intelligence (AI) and Data Science

Information and Communication Technologies (ICT)
- Unmanned systems (aerial, ground and marine)
- Biorobotics (bioinspired and medical)
- Autonomous manipulation
- Maching learning
- Image and video analysis and processing
- Data mining and big data analytics
- Efficient algorithms and optimization
- Biometrics and bio-imaging

- Next Generation Networks (NGNs) and NGN-enabled ICT applications and services
- Cyber Security technologies and applications
- Electronic and photonic circuits and devices
- Sensors

Security
- Security environments and security policies
- Defense science and technology